AI ASSISTED CODING
Assignment 3.1
2303A510B7
Batch - 14

Lab Experiment: Prompt Engineering – Improving Prompts and Context Management

Question 1: Zero-Shot Prompting (Palindrome Number Program)

Prompt:

Write a Python function that checks whether a given number is a palindrome and returns True or False.

Code & Output:

lab-3.1.py > ...

```python
#Generate a Python function that checks whether a given number is a palindrome
#and returns True or False
def is_palindrome_number(num):
    return str(num) == str(num)[::-1]
num = int(input("Enter a number : "))
print(f"{num} : {is_palindrome_number(num)}")
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number : 23432
23432 : True
PS C:\Users\danda\OneDrive\Documents\3.2\AIA>
```

Explanation:

- Define a function called is_palindrome_number that takes a single argument num.
- Convert the number to a string using str(num)
- Reverse the string using [::-1]
- Compare the original string with the reversed string using ==
- Return the result of the comparison
- Print the result of the function call

Question 2: One-Shot Prompting (Factorial Calculation)

Prompt:

Example: Input: 5 → Output: 120
Using this example, write a Python function to compute the factorial of a given number.

Code & Output :

lab-3.1.py > ...

```python
#Example: Input: 5 → Output: 120
#Using this example, write a Python function to compute the factorial of a given number.

def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n-1)

num = int(input("Enter a number: "))
print(f"The factorial of {num} is {factorial(num)}")
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 5
The factorial of 5 is 120
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 10
The factorial of 10 is 3628800
PS C:\Users\danda\OneDrive\Documents\3.2\AIA>
```

Explanation:

- Define a function called factorial that takes a single argument n.
- If n is 0, return 1
- Else, return n * factorial(n-1)
- Print the result of the function call

Question 3: Few-Shot Prompting (Armstrong Number Check)

Prompt:

Write a Python function to check whether a number is an Armstrong number.

Code & Output:

lab-3.1.py > ...

```python
# Write a Python function to check whether a number is an Armstrong number OR NOT using function.
def is_armstrong_number(num):
    sum = 0
    temp = num
    count = len(str(num))
    while temp > 0:
        digit = temp % 10
        sum += digit ** count
        temp //= 10
    return sum == num
num = int(input("Enter a number: "))
if is_armstrong_number(num):
    print(f"{num} is an Armstrong Number")
else:
    print(f"{num} is not an Armstrong Number")
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 1634
1634 is an Armstrong Number
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 370
370 is an Armstrong Number
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 123
123 is not an Armstrong Number
PS C:\Users\danda\OneDrive\Documents\3.2\AIA>
```

Explanation:

- Define a function called is_armstrong_number that takes a single argument num.
- Initialize a variable called sum to 0.

- Initialize a variable called temp to num.
- Initialize a variable called count to the length of the number.
- While temp is greater than 0, do the following:
- Get the last digit of temp using temp % 10.
- Add the cube of the last digit to sum.
- Divide temp by 10 to remove the last digit.
- Return True if sum is equal to num, otherwise return False.
- Print the result of the function call.

Question 4: Context-Managed Prompting (Optimized Number Classification)

Prompt:

Write an optimized Python program that classifies a number as prime, composite, or neither. Ensure proper input validation and efficient logic.

Code & Output:

lab-3.1.py > ...

```python
# Write an optimized Python program that classifies a number as prime, composite, or neither.
# Ensure proper input validation and efficient logic.
def classify_number(num):
    if num <= 1:
        return "Neither"
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return "Composite"
    return "Prime"
num = int(input("Enter a number: "))
print(f"{num} is {classify_number(num)}")
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 5
5 is Prime
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 6
6 is Composite
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 1
1 is Neither
PS C:\Users\danda\OneDrive\Documents\3.2\AIA>
```

Explanation:

- Define a function called classify_number that takes a single argument num.
- If num is less than or equal to 1, return "Neither".
- For i in range (2, int(num**0.5) + 1), do the following:
- If num is divisible by i, return "Composite".
- Return "Prime".
- Print the result of the function call.

Question 5: Zero-Shot Prompting (Perfect Number Check)

Prompt:

Write a Python function to check whether a given number is a perfect number.

Code & Output:

lab-3.1.py > ...

```python
#Write a Python function to check whether a given number is a perfect number
def is_perfect_number(num):
    sum = 0
    for i in range(1, num):
        if num % i == 0:
            sum += i
    return sum == num
num = int(input("Enter a number: "))
if is_perfect_number(num):
    print(f"{num} is a perfect number")
else:
    print(f"{num} is not a perfect number")
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 6
6 is a perfect number
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 28
28 is a perfect number
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 54
54 is not a perfect number
PS C:\Users\danda\OneDrive\Documents\3.2\AIA>
```

Explanation:

- Define a function called is_perfect_number that takes a single argument num.
- Initialize a variable called sum to 0.
- For i in range (1, num), do the following:
- If num is divisible by i, add i to sum.
- Return True if sum is equal to num, otherwise return False.
- Print the result of the function call.

Question 6: Few-Shot Prompting (Even or Odd Classification with Validation)

Prompt:

Write a Python program to classify a number as even or odd with proper validation.

Code & Output:

lab-3.1.py > ...

```python
#write a Python program to classify a number as even or odd with proper validation.
def classify_number(num):
    if num % 2 == 0:
        return "Even"
    else:
        return "Odd"
num = int(input("Enter a number: "))
print(f"{num} is {classify_number(num)}")
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 8
8 is Even
PS C:\Users\danda\OneDrive\Documents\3.2\AIA> & C:\Users\danda\AppData\Local\Programs\Python\Python313\python.exe
 c:/Users/danda/OneDrive/Documents/3.2/AIA/lab-3.1.py
Enter a number: 9
9 is Odd
PS C:\Users\danda\OneDrive\Documents\3.2\AIA>
```

Explanation:

- Define a function called classify_number that takes a single argument num.
- If num is divisible by 2, return "Even".
- Else, return "Odd".
- Print the result of the function call.
- Print the number and the result of the function call.